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END TERM EXAMINATION
SECOND SEMESTER [B.TECH] JUNE 2024

Paper Code: BS-112

Subject: Applied Mathematics-II

Time: 3 Hours

Maximum Marks: 60

Note: Attempt any five questions including Q.No 1 which is compulsory. Select one question from each unit.

Q. No.	Question	Marks
Q1	Attempt all of the following:	(3×4=12)
(a)	Resolve $e^{\sin(x + iy)}$ into real and imaginary parts.	(3)
(b)	Find the image of the infinite strip $1/4 \leq y \leq 1/2$ under the transformation $w = 1/z$.	(3)
(c)	Find the Laplace transform of $(\cos at - \cos bt) / t$.	(3)
(d)	Using the method of Separation of variables, solve $\partial u / \partial x = 2(\partial u / \partial t) + u$, where $u(x, 0) = 6e^{-3x}$.	(3)
UNIT - I		
Q2 (a)	Find the value of C_1 and C_2 such that the function $f(z) = x^2 + C_1 y^2 - 2xy + i(C_2 x^2 - y^2 + 2xy)$ is analytic. Also find $f'(z)$.	(6)
(b)	Show that the function $u = (1/2) \ln(x^2 + y^2)$ is harmonic. Find its harmonic conjugate.	(6)
OR		
Q3 (a)	State Cauchy Integral formula and hence evaluate $\oint_C [(3z^2 + z) / (z^2 - 1)] dz$, where C is the circle $ z - 1 = 1$.	(6)
(b)	Evaluate the Line Integral $\int_C z^2 dz$, where C is the boundary of a triangle with vertices $0, 0 + i, -1 + i$, traversed clockwise.	(6)
UNIT - II		
Q4 (a)	Evaluate $\int_C [z^2 / ((z - 1)^2(z + 2))] dz$, where C is $ z = 3$ by using Cauchy residue theorem.	(6)
(b)	Evaluate the Laurent series which represents the function $f(z) = (z^2 - 1) / ((z + 2)(z + 3))$ when: (i) $ z < 2$ (ii) $2 < z < 3$	(6)
OR		
Q5 (a)	Apply the calculus of residues to evaluate $\int_{-\infty}^{\infty} [\cos x / ((x^2 + a^2)(x^2 + b^2))] dx$, where $a > b > 0$.	(6)
(b)	Let $f(z)$ be a bilinear transformation such that $f(\infty) = 1, f(i) = i, f(-i) = -i$. Find the image of the unit disk $C = \{z \in \mathbb{C} : z < 1\}$ under $f(z)$.	(6)

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Q. No.	Question	Marks
UNIT - III		
Q6 (a)	Find the Inverse Laplace transform of $F(s) = \log[(s + a) / (s + b)]$.	(6)
(b)	Find the Fourier series for: $f(x) = -\pi$ for $-\pi < x < 0$, $f(x) = x$ for $0 < x < \pi$. And deduce that $1/1^2 + 1/3^2 + 1/5^2 + \dots = \pi^2 / 8$.	(6)
OR		
Q7 (a)	Solve $d^3y/dt^3 + 2(d^2y/dt^2) - dy/dt - 2y = 0$, given $y = dy/dt = 0$ and $d^2y/dt^2 = 6$ at $t = 0$ by using Laplace transformation.	(6)
(b)	Find the Fourier Transform of: $f(x) = 1$ for $ x < 1$, and 0 for $ x > 1$. Hence evaluate $\int_0^\infty [(\sin x) / x] dx$.	(6)
UNIT - IV		
Q8 (a)	A string is stretched and fastened to two points l apart. Motion is started by displacing the string in the form $y = a \sin(\pi x / l)$ from which it is released at time $t = 0$. Show that the displacement of any point at distance x from one end at time t is given by: $y(x, t) = a \sin(\pi x / l) \cos(\pi c t / l)$.	(6)
(b)	Find the temperature in a bar of length 2 whose ends are kept at zero and lateral surface insulated if the initial temperature is $\sin(\pi x / 2) + 3 \sin(5\pi x / 2)$.	(6)
OR		
Q9 (a)	An infinitely long plane uniform plate is bounded by two parallel edges and an end at right angles to them. The breadth is π . This end is maintained at temperature u_0 at all points and the other edges are at zero temperature. Determine the temperature at any point of the plate in the steady state.	(6)
(b)	Solve $(1/4) u_{xx} = u_{tt}$ with initial condition $u(x, 0) = 0$, $u_t(x, 0) = 8 \sin 2x$ using D'Alembert's Principle.	(6)

**** End of Question Paper ****